Name: _____ Date: _____ Period: _____

3-2, Day 2: Splitting the Middle

TEKS P.5.J: solve polynomial equations by applying a variety of techniques. Factoring is the tool. Tags: **B** = Basic (leading coefficient 1) · **I** = Intermediate (leading coefficient greater than 1).

Warm-up: What you already know

Factor each. Check by multiplying.

1. $x^2 + 7x + 12$ B

2. $x^2 - 2x - 15$ B

3. By grouping: $6x^2 + 9x + 4x + 6$ B

Part A: The ac method, step by step

Factor **$6x^2 + 11x + 4$**. Fill in each step.

Step	Your work
1. Find a, b, c. Then a · c.	a = _____ b = _____ c = _____ a · c = _____
2. Find two numbers with product a · c and sum b.	_____ and _____ (list the factor pairs you tried)
3. Split the middle term.	$6x^2$ + _____x + _____x + 4
4. Group, and factor each pair.	($6x^2$ + _____x) + (_____x + 4) = _____ (_____) + _____ (_____)
5. Factor out the matching binomial.	(_____)(_____)
6. Check: multiply back.	Do you get $6x^2 + 11x + 4$?

Look at the two numbers you found in step 2. How do they connect to the 6, the 4, and the 11?

_____

_____

Name: _____ Date: _____ Period: _____

Part B: A different route, the box

Use the same trinomial, **$6x^2 + 11x + 4$**, and the two numbers from Part A.

	Label: _____	**Label: _____**
Label: _____	$6x^2$	
Label: _____		4

(a) Put your two numbers (as x-terms) in the empty cells. Label each row and column with what its two cells share.

(b) The factors are (_____)(_____).

(c) Swap where the two x-terms go. What changes? What stays the same?

_____

Practice

Factor completely. Any correct method counts (ac method, box, trial and check). Take out a GCF first if there is one, and check by multiplying back. Use scratch paper if you need more room.

4. $2x^2 + 7x + 3$ **I**

5. $3x^2 - 11x + 6$ **I**

6. $2x^2 + x - 6$ **I**

7. $6x^2 - 5x - 4$ **I**

8. $9x^2 - 12x + 4$ **I**

9. $6x^2 + 21x + 9$ (What do all three terms share?) **I**

Name: _____ Date: _____ Period: _____

10. $8x^2 - 20x - 12$ I

11. $2x^2 + 3x + 4$ (If it is prime, show how you know.) I

Problems 12–14: solve by factoring. Give exact answers (fractions are fine). Check one solution by substituting.

12. $2x^2 + 5x + 2 = 0$ I

13. $6x^2 = 7x + 5$ I

14. $9x^2 + 3x - 12 = 0$ I

15. A rectangular patio has area $\mathbf{A = 6x^2 + 17x + 5}$ square feet, where x is a design measurement in feet. I

(a) Factor to write the length and width as expressions in x.

(b) If $x = 2$, what are the length, width, and area?

(c) The patio must have an area of 15 square feet. Solve $6x^2 + 17x + 5 = 15$. Which solution(s) make sense for the patio? Why?

Math Time Machine: Around 628, Brahmagupta in India gave one of the first general rules for solving quadratics and treated zero and negative numbers as real quantities. Around 820, al-Khwarizmi's book (its title includes *al-jabr*) gave us the word **algebra**. The old Chinese *Nine Chapters* uses rectangles and areas, a cousin of your box.